MA203-Tutorial-Module I

1. (a) Find the numbers of significant figures in 1.0100.
   (b) Find the relative error of the number 8.6 if both the digits are correct.

2. Round off the number 865250 and 37.46235 to four significant figures and compute absolute error, relative error and percentage error.

3. Evaluate the sum S= $\sqrt{3} + \sqrt{5} + \sqrt{7}$ to 4 significant digits and find its absolute and relative errors

4. Find a suitable alternative representation of the equation $f(x) = x^3 + x^2 - 1 = 0$, in the form $x = \varphi(x)$ for finding its root in the interval [0, 3] by iterative method (fixed point method).

5. Find the absolute difference between 4th and 5th approximation to the real root of the equation $(x) = x^3 - 5x - 7 = 0$ using method of false position correct upto 4th decimal places.

6. The real sequence generated by iterative scheme $x_n = \frac{x_{n-1}}{2} + \frac{1}{x_{n-1}}, n \geq 1$
   (a) converges to $\sqrt{2}$ for all $x_0 \in \mathbb{R} \setminus \{0\}$
   (b) converges to $\sqrt{2}$ whenever $x_0 > \sqrt{2/3}$
   (c) converges to $\sqrt{2}$ whenever $x_0 \in (-1,1) \setminus \{0\}$
   (d) diverges for any $x_0 \neq 0$

7. One root of the equation lies in the interval (3,4). Find the least number of iterations needed for the bisection method so that $|error| \leq 10^{-3}$

8. What is the rate of convergence of the Newton-Raphson method?

9. Find a real root of the equation $f(x) = x^3 - 5x + 1 = 0$ in the interval (0,1). using Secant method after four iterations (take $x_0 = 0, x_1 = 1$).

10. Suppose the transcendental equation is $xe^x = 1$ and the root lies in the interval [0,1]. Find the approximate value of the root correct to 2 decimal places.

11. Find the value of $\sqrt{10}$ using Newton-Raphson method with initial guess 2 correct upto 4 decimal places.

12. Find a root of the equation $cosx - xe^x = 0$ by Regula Falsi method correct to 4 decimal places.

13. The smallest positive root of the equation $f(x) = x^4 - 3x^2 + x - 10 = 0$ is to be obtained
    (a) find an interval of unit length which contains this root
    (b) Perform two iterations of the bisection method

14. What should be the value of k so that the iteration formula $x = x + k(x^2 - 3)$ may converge at a good rate, given that x = $\sqrt{3}$ is a root

15. Consider the series $x_{n+1} = \frac{x_n}{2} + \frac{9}{8x_n}$, $x_0 = 0.5$ obtained from the Newton-Raphson method. Show that the series converges to 1.5